

Crowd-sourcing a benchmark for English forced Alignment

Michael McAuliffe & Kaylynn Gunter

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<https://splaot-lab.github.io/mon-align>

Schedule

- Brief introduction (10 minutes)
 - Goals of the workshop
 - Data and procedure
 - General philosophy and approach to segmentation
- Overview of the annotation conventions (15 minutes)
- TextGrid correction! (2.5 hours)
 - We'll have 2 ~10 min breaks
- Wrap up (20 mins)
 - Returning data/uploading
 - Survey

Goals

- Creating an open-source, easily distributable benchmark for forced alignment
 - Permissible license
 - Represents “easy” to segment English
 - Not spontaneous speech
 - Distributed on Hugging Face as a benchmark evaluation for forced alignment
- You might ask, doesn't this kind of dataset already exist?

Who needs more English datasets?

- Current manually-aligned datasets for English (that we're aware of) are:
 - TIMIT (Small, costs money, can't be distributed, idiosyncratic format)
 - Buckeye (Good size, free, but can't be distributed, idiosyncratic format)
- Additional issues
 - Errors in the original need fixing by every user – meaning variable ground truth across researchers, not a single benchmark
 - Narrow transcription
 - Issues in evaluation
 - Decreased inter-annotator agreement
 - But no metadata on the annotator for a given file

Goals

- Evaluating the evaluation data
 - Studying disagreement in segmentation
- Segmentation is an analysis by the annotator
 - Segmentation is subjective, to different degrees; there is no objective ground truth segmentation
 - There are cases of convergence in analysis between annotators, and others where there is greater variability across annotators.

Goals

- In benchmark evaluation, errors are weighted equally
 - Boundary between a stop and a vowel (*toe* /T OW1/
 - Short transition
 - Trained phoneticians/conventions will largely agree within small tolerance (<10 ms)
 - Boundary between a vowel and a liquid (*all* /AO1 L/
 - Longer transition
 - Trained phoneticians/conventions may disagree (beginning, end, midpoint of transition)
- “Easy” boundaries should penalize automatic systems more

Goals

- Create better tooling!
- Praat is the standard, but has limitations
 - Manual correction takes a lot of time
 - Requires human intervention at every stage, prone to error
- Today, we're using a Praat script
 - Handles loading files, autosaving, navigation, shortcuts for insertion/deletion
- But: Feedback for annotation tooling
 - We can create more dedicated tooling that targets the use case of correcting segmentation
 - Survey at end will elicit feedback on your experience today

Data and Procedure

- We want to annotate (generally) easy files today
- LibriSpeech “clean” test set (2.6k files)
 - Read speech
 - 40 speakers
 - 8 minutes of audio per speaker
 - Low Word Error Rate for most ASR systems
- Data for today (1.7k files)
 - < 10 seconds (614 files)
 - < 14 phones per second (335 files)
 - No audio peaking or loud transients (12 files)
 - Excluded files are in “extras” folder and can be annotated, but won’t likely have time today

Data and Procedure

- Every annotator has:
 - 5 “cohort” files
 - This set of file is shared with 4 other annotators (those at your table)
 - Should have a good distribution of trickier cases to consult on
 - Use this set to calibrate yourself to the conventions and each other
 - 3 “common” files
 - This set is shared with all other annotators
 - 41 “individual” files
 - Only for this annotator
 - No selection criteria other than not being excluded
 - 24 “extra” files
 - Excluded files and leftovers from sampling

Procedure

- Praat script has a selection for which subset to annotate
- First, run with “cohort” files
 - Work together with your cohort for any questions, uncertainties, conventions
 - We can update conventions as needed with problematic examples
- Then run with “common” files
 - Work together with your cohort for any questions, uncertainties, conventions
- Then run with “individual” files
 - Won't be shared with your cohort, but can still ask questions when you get stuck
- (Probably won't be able to) run with “extra” files

Guiding Principles

- What we want is a “broadly accurate” segmentation
 - No allophonic labels
 - Removing segments that have been deleted/reduced
 - “Probably” /P R AA1 B AH0 B L IY0/ -> [P R AA1 B L IY0]
 - Unreleased stops are trickier, see annotation guidelines
- Goal is to balance speed with accuracy
 - Easy to get bogged down with single boundaries that defy conventions/analysis
 - There is typically no “right” answer
 - Annotating 10 easier boundaries is better than 1 speech error/reduced boundary
 - Just note in the README.md which files are problematic, adjust boundaries to be good enough

Guiding Principles

- Focus on the time alignment and not the phone labels
 - Unless adding or removing segments, in which case you will be asked to adjust the boundaries
- Using ARPAbet output from MFA
 - Broad US English phone set
 - Some pronunciations (mostly proper nouns) generated with G2P
 - Only correct phone labels if the pronunciations are egregiously wrong
- Things not to correct
 - Word boundaries
 - Stress numbers on vowels
 - Incorrect/reduced vowel quality

Remember!

- There is no such thing as perfectly segmented speech!
 - If you hit a point of “eh, good enough”, then that’s good enough for our purposes
- We are ok with variation in judgment/applying conventions
 - The cohort files should help calibrate judgements, but disagreements are still likely!
- This is a collaborative exercise
 - Feel free to ask your cohort members and us for second opinions!
- Have fun!
 - Speech is messy and audiobook narrators say the darnedest things!

Overview of annotation guidelines

Wrap up

Wrap up

- Please fill out the survey:
 - [insert link]
 - Link also in “README.md” in the annotation
- Uploading data
 - Please return any USBs to us!
 - Feel free to continue correcting your subset
 - Zero obligation or expectation!
 - If you do upload it, please let us know

Thank you so much for spending your afternoon annotating with us!